AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

BUS102
Introduction to Statistics

MTA Examination / Version: B
2025-2026 / Fall

15-November-2025

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed: 1.5 Hours**

Instructions:

Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
This is a **CLOSED** book examination. NO documents of any kind are allowed.
You **MUST** answer the questions using only the answer booklets provided
You **MUST** return the examination paper together with your answer booklets.
Do **NOT** write your answers in pencil except graphs/sketches
Students are **NOT** allowed to use any type of dictionary or mobile phones during the exa
Students are allowed to use ***non-programmable*** calculators only (if needed).
Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Poli

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Question 1: (10 marks)

1. Which of the following is an example of a categorical variable?

A) Weight
B) Voltage
C) Eye Color
D) Number of Children

2. What type of variable is 'Number of Children'?

A) Nominal
B) Continuous
C) Discrete
D) Ordinal

3. Which error is due to the respondent misunderstanding the question?

A) Sampling error
B) Nonresponse error
C) Coverage error
D) Measurement error

4. Which is a correct statement about cumulative frequency?

A) It's the difference between class limits
B) It's used only for categorical data
C) It adds up frequencies sequentially
D) It must equal 50% always

5. Which quartile divides the lowest 50% of the data?

A) Q1
B) Q2
C) Q3
D) Q4

6. What does the coefficient of variation measure?

A) Spread in absolute terms
B) Central tendency
C) Relative variation
D) Standardized mean

7. What is the complement of the event 'rolling a 6 on a die'?

A) Rolling a 1
B) Rolling a number less than or equal 6
C) Rolling a number other than 6
D) Rolling a 5

8. An event with a 100% probability is called:

A) Impossible event
B) Certain event
C) Likely event
D) Rare event

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9. Which of the following cannot be a probability?

A) 0.75 B) 1

C) -0.2 D) 0

10. If the probability of passing an exam is 0.85, what is the probability of failing?

A) 0.25 B) 0.15

C) 0.85 D) 1.15

Question 2: (10 marks)

A. The following table shows the frequency distribution of hours studied by a group of students during a week. (5 marks)

Hours Studied	Frequency	Relative Frequency	Cumulative Frequency	Cumulative %
0 to 5	3			
5 to 10	7			
10 to 15	9			
15 to 20	6			
20 to 25	5			

Calculate the following:

a. Relative frequency
b. Cumulative frequency
c. Cumulative frequency percentage.

B. Customers gave the following ratings (on a scale of 1 to 10) to a new product: 5, 6, 6, 7, 8, 9, and 10.
What are the mean, median, and mode of the customer ratings? (5 marks)

Question 3: (10 marks)

A. A machine produced the following units over five days: 50, 52, 48, 49, and 51. Calculate the sample variance, sample standard deviation, and coefficient of variation of daily output. (5 marks)

B. The following Table represents the summary for a survey that taken from 270 persons about their habits. (5 marks)

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Age Group	Reading	Gaming	Sports	Total
Teenagers	35	45	40	120
Adults	50	30	20	100
Seniors	30	10	10	50
Total	115	85	70	270

Find:

a) What is the probability that a person prefers Reading given they are a Senior?
b) What is the probability that a randomly selected person prefers Sports?
c) What is the probability that a randomly selected person is a Senior and prefers Gaming?
d) What is the probability that a person prefers Gaming or is a Teenager?
e) What is the probability that a randomly selected an adult person?

End of questions

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FORMULAS

Sample Mean

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n} \quad (3.1)$$

Median

$$\text{Median} = \frac{n+1}{2} \text{ ranked value} \quad (3.2)$$

Geometric Mean

$$\overline{X}_G = (X_1 \times X_2 \times \cdots \times X_n)^{1/n} \quad (3.3)$$

Geometric Mean Rate of Return

$$\overline{R}_G = [(1 + R_1) \times (1 + R_2) \times \cdots \times (1 + R_n)]^{1/n} - 1 \quad (3.4)$$

Range

$$\text{Range} = X_{\text{largest}} - X_{\text{smallest}} \quad (3.5)$$

Sample Variance

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1} \quad (3.6)$$

Sample Standard Deviation

$$S = \sqrt{S^2} = \sqrt{\frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}} \quad (3.7)$$

Coefficient of Variation

$$CV = \left(\frac{S}{\overline{X}}\right)100\% \quad (3.8)$$

Z Score

$$Z = \frac{X - \overline{X}}{S} \quad (3.9)$$

First Quartile, Q_1

$$Q_1 = \frac{n+1}{4} \text{ ranked value}$$

Third Quartile, Q_3

$$Q_3 = \frac{3(n+1)}{4} \text{ ranked value}$$

Interquartile Range

$$\text{Interquartile range} = Q_3 - Q_1$$

Population Mean

$$\mu = \frac{\sum_{i=1}^{N} X_i}{N}$$

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^{N} (X_i - \mu)^2}{N}$$

Population Standard Deviation

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (X_i - \mu)^2}{N}}$$

Sample Covariance

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})}{n-1}$$

Sample Coefficient of Correlation

$$r = \frac{\text{cov}(X, Y)}{S_X S_Y}$$

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Probability of Occurrence

$$\text{Probability of occurrence} = \frac{X}{T} \tag{4.1}$$

Marginal Probability

$$P(A) = P(A \text{ and } B_1) + P(A \text{ and } B_2) + \cdots + P(A \text{ and } B_k) \tag{4.2}$$

General Addition Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \tag{4.3}$$

Conditional Probability

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} \tag{4.4a}$$

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)} \tag{4.4b}$$

Independence

$$P(A|B) = P(A) \tag{4.5}$$

General Multiplication Rule

$$P(A \text{ and } B) = P(A|B)P(B) \tag{4.6}$$

Multiplication Rule for Independent Events

$$P(A \text{ and } B) = P(A)P(B) \tag{4.7}$$

Marginal Probability Using the General Multiplication Rule

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_k)P(B_k)$$

Bayes' Theorem

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_k)P(B_k)}$$

Counting Rule 1

k^n

Counting Rule 2

$(k_1)(k_2)\ldots(k_n)$

Counting Rule 3

$n! = (n)(n-1)\ldots(1)$

Counting Rule 4: Permutations

$${}_nP_x = \frac{n!}{(n-x)!}$$

Counting Rule 5: Combinations

$${}_nC_x = \frac{n!}{x!(n-x)!}$$

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